Gleipnir: Frontier-Quality Control Monitors at a Fraction of the Cost

Sprint proposal: from building the monitors to stress-testing them

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The problem: monitoring imposes a safety tax

- Agent harnesses now score each action with a monitor *before* it executes.
- This puts the monitor on the critical path: its latency adds to every action and its compute scales with the deployment.
- Frontier models monitor well, but they are too slow and expensive to run on every action.

Goal

Small, open-weight monitors with frontier-level quality that are cheap enough to run on every action and open enough for anyone to study and modify.

Core idea: ask for one decision and distill its logits

Prior work: distill rationales

Teacher writes a rationale and a 0–10 score.

Student learns to reproduce the text (SFT), then RL.

Every evaluation **generates hundreds of tokens**.

Gleipnir: distill the decision

Ask the teacher (Kimi K3) for a single token: 0 or 1.

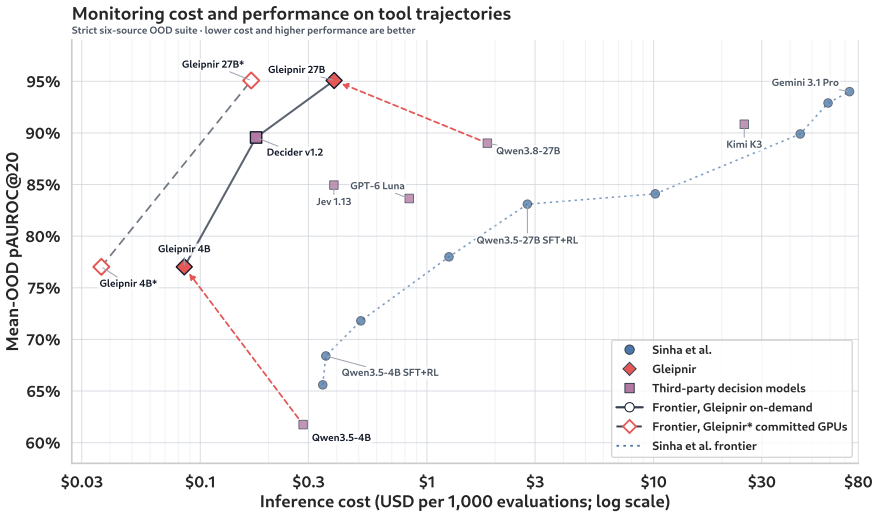
The student matches the teacher's **probability** on that token (soft logit targets, LoRA).

Every evaluation is **one prefill**; the score is a calibrated probability.

No rationale targets and no RL. The teacher's uncertainty carries the information that rationales would otherwise have to carry.

Sinha et al. (2026), *Training Deliberative Monitors for Black-Box Scheming Detection*.

Where we stand on the cost–performance frontier



6,395 OOD trajectories from six sources (Sinha et al.). Gleipnir at self-hosted B200 break-even; * committed GPUs.

About 200× cheaper than the prior frontier, at equal quality

Monitor	Mean-OOD pAUROC@20	$ / 1K evaluations
Gemini 3.1 Pro, prompted (prior frontier top)	0.940	73.22
Claude Opus 4.6, prompted	0.942	94.79
Gleipnir 27B	**0.951**	**0.39** (0.17*)
Qwen3.5-4B SFT+RL (Sinha et al.)	0.684	0.36
Gleipnir 4B	**0.770**	**0.085** (0.037*)

Gleipnir 27B beats Gemini 3.1 Pro at **1/188 of the cost** on-demand (1/437 with committed GPUs).

* Committed-use B200 rate ($3.44/h); otherwise on-demand ($7.99/h). The cost is inference only and excludes training.

Status: ahead of the proposal

Proposal, week 1: improve the monitors

- Done. Gleipnir 4B (seed average + quantization-aware distillation) and Gleipnir 27B are frozen release candidates.
- Optimized B200 serving and release packaging are built.

Proposal, week 2: stress-test them

- Starting now: the whole sprint goes to the second half of the proposal.
- Both monitors are frozen, so every result describes the release checkpoints.

Sprint deliverable

A **model card** for each monitor that tells the community what it is good at, where it fails and how easily it can be fooled.

Four avenues for the model card

1. Fresh benchmarks

One-shot evaluation of the frozen monitors on benchmarks never used in the project (MonitoringBench, SleightBench).

Low effort

2. Black-box vs. white-box monitors

Compare against probes. We know of no benchmark with *open-model* agents on control tasks; building one is slow.

High effort

3. Attacks and mechanistic fixes

The weights are public, so white-box attacks on one decision logit are easy to find. How easily does a decision flip? Can ablating directions remove unwanted traits?

Medium effort

4. Newly affordable domains

Settings that were too costly to monitor before, such as agent swarms: one million actions cost $390 with Gleipnir 27B compared with $73K with Gemini 3.1 Pro.

Medium effort

Where we would like feedback

- **Priorities.** Which of the four avenues would make the model card most useful to the control community in two weeks?
- **Benchmarks.** Beyond MonitoringBench and SleightBench, which results would be most convincing?
- **White-box comparison.** Is there existing data with open-model agents on control tasks, or is building our own worth the time?
- **Attacks.** Which attack settings would be credible: white-box only, or black-box query attacks as for an API-served monitor too?
- **New domains.** Which setting would cheap monitoring unlock that matters most: swarms, long-horizon sessions, or per-token monitoring?